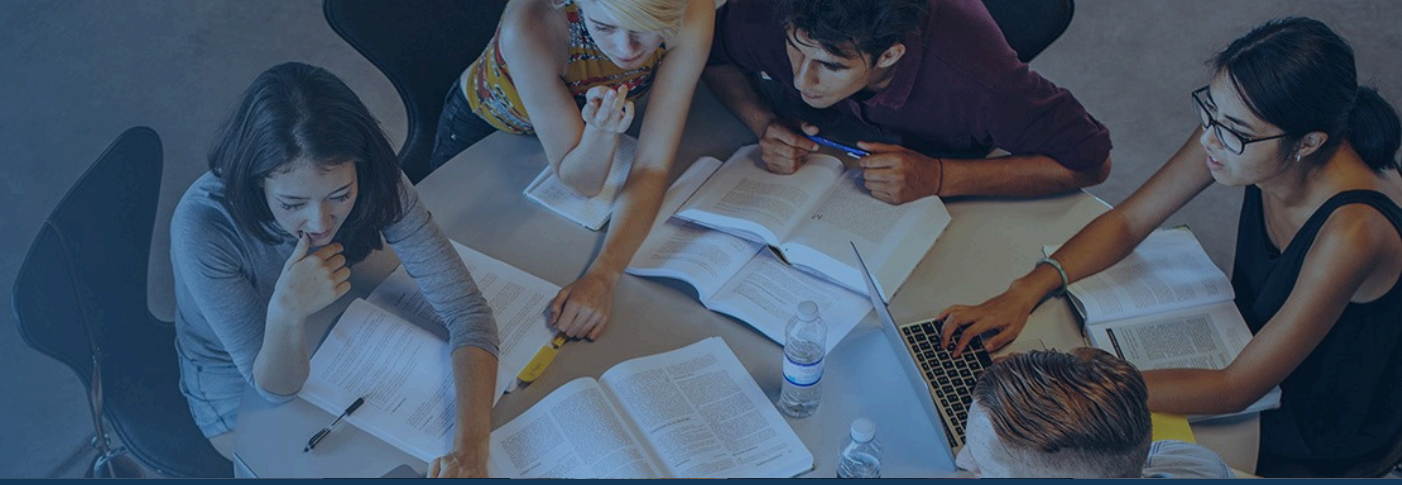
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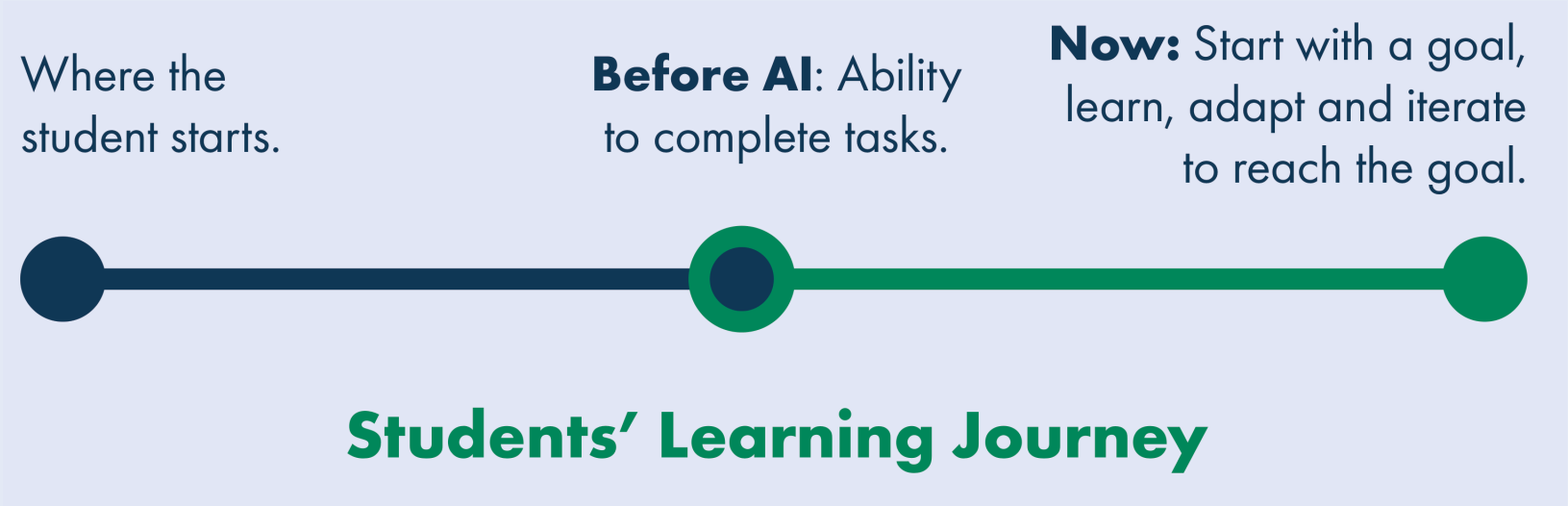
Learn. Connect. Solve.

Human Value that Grows with AI Capability

Building and testing post-secondary learning experiences for the age of AI

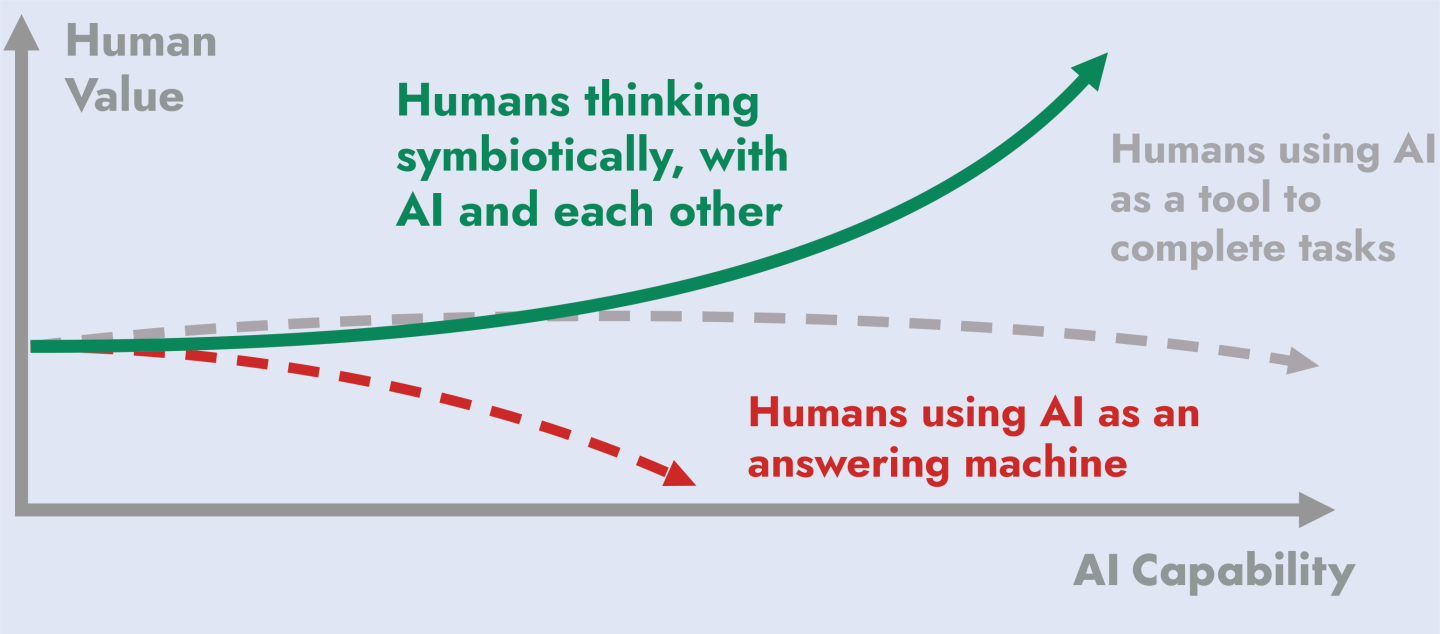
The Challenge

The expectations for new graduates are shifting to be more like those of someone with years of experience, who can pursue a complex goal.



Our Hypothesis

Human value can grow with AI capability if students learn to think with AI and each other.



Jagged frontier: Dell'Acqua, F. et al. *Navigating the Jagged Technological Frontier.* Harvard Business School Working Paper 24-013, 2023.

Our Approach

We believe three things are needed to give the large middle of society the opportunity to thrive through the AI transition. We are working on all three.



A precise, actionable definition of the problem

We want students to move from completing assigned tasks to pursuing long-term, complex goals with AI.

A well-defined, teachable framework of capabilities

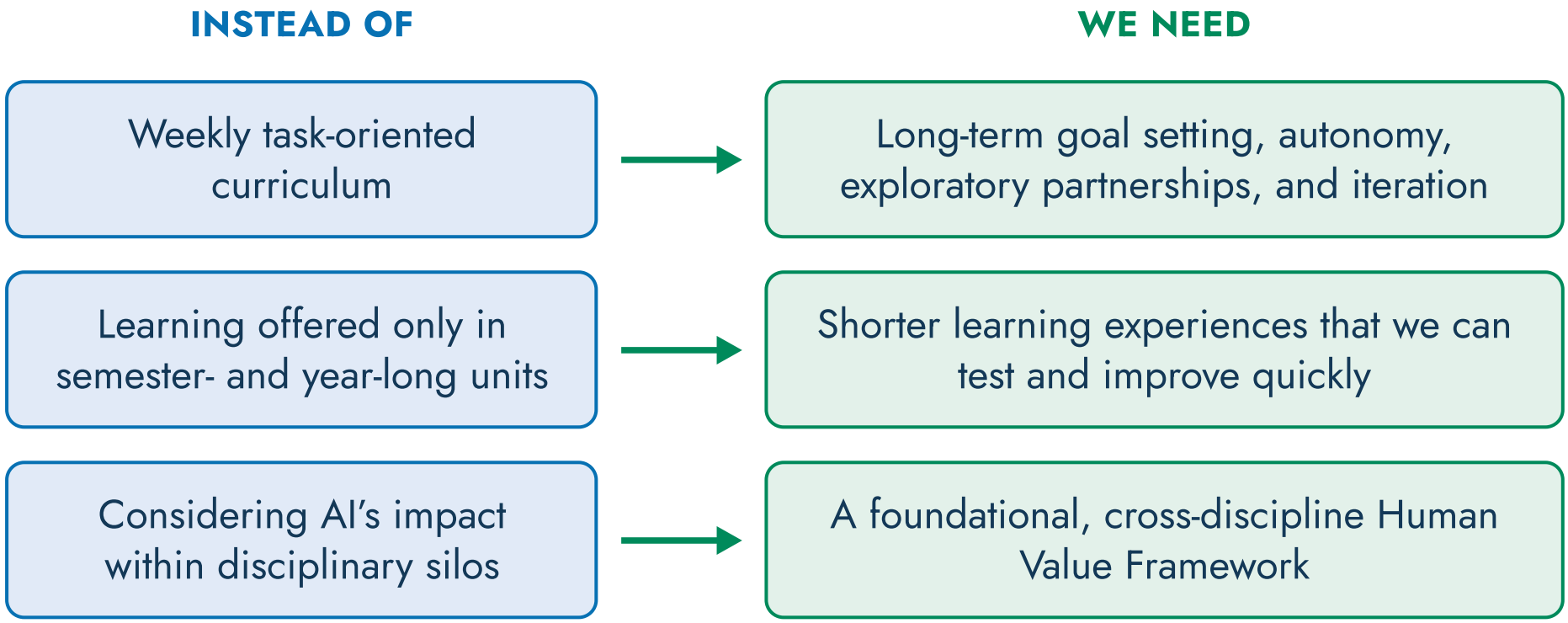
We describe the capabilities students need and how we think they can be built.

A rigorous, adaptive experimental process

We develop our content and test it with students at community colleges and CSUs, so that it stays aligned with their challenges and opportunities.

Transformation is Needed

We need to rethink structural limitations in post-secondary learning and approach the AI transition differently.



Deci, E. L. and Ryan, R. M. Self-determination theory: people grow when they experience autonomy, connectedness and competence.
Lerner, J. S., and Tetlock, P. E. (1999). Accounting for the effects of accountability. *Psychological Bulletin*, 125(2), 255–275.

Our Proposed Human Value Framework

Symbiotic Thinking: The human-led practice of pursuing wisdom in partnership with other intelligences, human or artificial.

LEADING TO THREE CAPABILITIES



LEADING TO TWO OUTCOMES

Superagency & Human Value

Build · Measure · Learn

We are testing the Human Value Framework with different audiences.

Applying AI at Work Certificate with De Anza Community College

Working professionals and career changers take a real problem from their work or personal life and ask good questions to frame it clearly, find its root cause and build an effective solution with AI.

Career Intelligence Workshop

Seniors and recent graduates learn to understand the market, reflect on their interests and evaluate their career readiness, so that they can identify their niche and take ownership of their job search.

Multiple courses at Cal State Monterey Bay

Freshmen through seniors, in courses ranging from general education physics to Capstone, practice the three capabilities through conversations with peers and AI (Symbiotic Thinking) to build Superagency and understand their Human Value.

Our own learning also relies primarily on conversations. We talk with learners 1-on-1 and in small groups, read the chat transcripts from our customized AI Dojo, and read responses to the guided activities we have designed.

Build, measure, learn: Ries, E. *The Lean Startup*, 2011.

Early Signals We Track

Right now, we are tracking these six qualities as the best early signals for nurturing the three capabilities of self-directed learning, integrative solving and adaptive building.

INWARD FACING

- Self-knowledge
- Self-regulation
- Owning the outcome

OUTWARD FACING

- Initiative
- Working with uncertainty
- Adaptability

We look for evidence of these six qualities in the work students submit to meet the course learning outcomes.

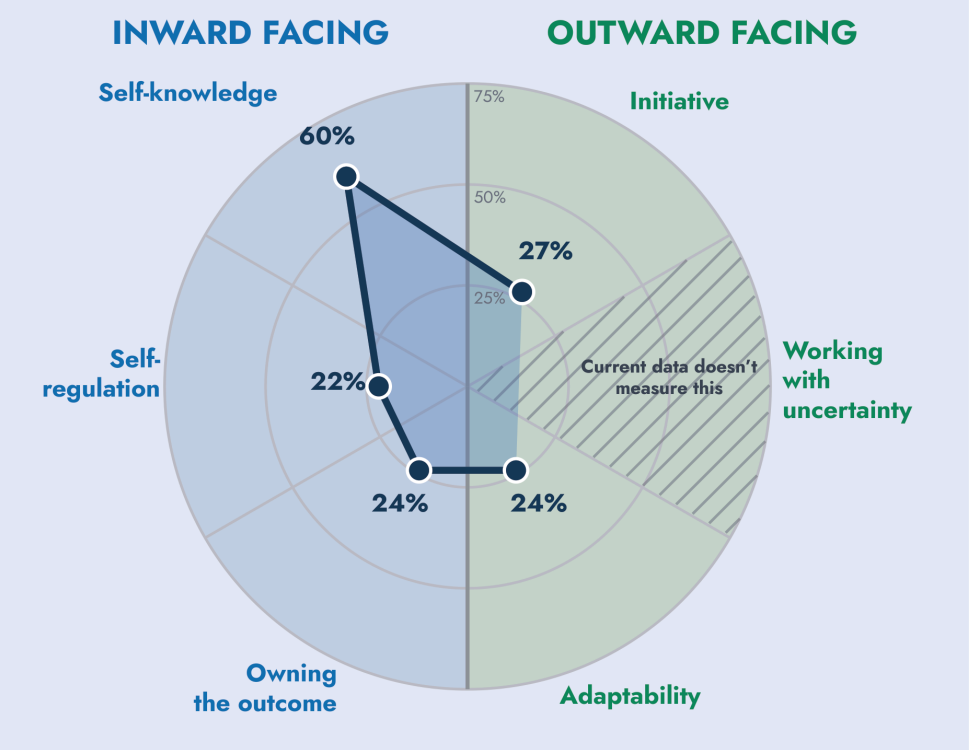
How a rating is made. Selected assignments are evaluated against a rubric to identify evidence of one or more of these qualities.

Early Data We Have Seen

This chart shows what we have seen so far in tracking these qualities in students:

- 129 students in two Cal State Monterey Bay courses (first-year and junior)
- Each percentage is the share where we saw signals for the quality twice: in take-home reflections and in a proctored, handwritten exam

This chart demonstrates the potential of AI analysis to track students' growth on the six qualities, based on their work in the course. The analysis shown was done primarily by AI on real student work, and human review is in progress.



We are developing a rigorous human-in-the-loop process to nurture and track these qualities.

Continuing Work

There is a lot we do not know. We are excited about building on our current work:

Learning by watching and reading. We follow the work in this space carefully, from the research on how people learn with AI to what other programs are trying, and we use it to change our own thinking.

Learning through doing. We experiment in our own work and in the learning experiences we offer students to better understand AI and its impact. We use short experiments as early signals, allowing us to test and refine the framework through quick iterations.

Learning through building. We are building the AI Dojo and The Commons as two tools that make AI a support for Symbiotic Thinking and human connection: **dojo.symbioticthinking.ai** and **the-commons.symbioticthinking.ai**

Learning from other disciplines. We are working closely with faculty from Kinesiology. We are also extending our open-source work-based learning experience program to serve students from other disciplines.

Learning through collaboration. Our course pages, assignment designs and analysis are publicly available while the courses run. We believe transparency and collaboration will help us all learn and improve more quickly.

profsathya.github.io/Common-Curriculum/home.html

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dojo.symbioticthinking.ai/cti

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